

EDUCATION	
Birla Institute of Technology Mesra	Ranchi, India
Integrated Master's of Science - Mathematics & Computing; GPA: 8.88	2019 - 2024

PUBLICATIONS	
Transformers Remember First, Forget Last: Dual-Process Interference in LLMs arXiv 2025. K-PERM: Personalized Response Generation Using Dynamic Knowledge Retrieval and Persona-Adaptive Queries at AAAI Spring Symposium 2024. Multimodal Ingredient Substitution Knowledge Graph (MISKG) Dataset Release 2024. Cook-Gen: Robust Generative Modeling of Cooking Actions from Recipes at IEEE SMC 2023. Spatial Field Fusion Network (SFFNet) for Panoramic Dental X-ray Segmentation at IEEE APSCON 2023.	

EXPERIENCE	
Thomson Reuters Lab	Bangalore, India
Applied Research Scientist	Aug 2024 - Present
◦ LLM Reasoning & Reward Modeling: * Fine-tuned Qwen3.5-9B via Dr. GRPO for multi-turn legal citation retrieval (1,139 Swiss law queries): designed 3-tool agentic environment (BM25 search, article verification, submit) through 29-experiment ablation — 7-tool design collapsed to F1 0.066 vs. F1 0.244 with 3 tools * Designed multi-stage reward: bootstrap stage rewards search behavior and citation format; optimize stage uses F1 with query quality, anti-hallucination, and over-prediction penalties * Built IRAC (Issue-Rule-Application-Conclusion) Knowledge Graph from Westlaw corpus for DPO fine-tuning preference data; improved LegalBench benchmark performance ◦ Test-Time Scaling: * Implemented production module supporting Best-of-N, DVTS, Beam Search, and Particle Filtering steered by fine-tuned Bradley-Terry reward models (outcome and process) * Conducting first systematic study of test-time scaling on non-verifiable tasks (targeting NeurIPS 2026): compute-matched (inference-FLOPs) benchmarking of 6 TTS paradigms across 5 open-ended QA domains (legal, medical, finance, instruction-following, writing); ~8,500 samples spanning deterministic to rubric-only verifiability ◦ Agentic Systems: * Architected multi-agent system for automated legal document editing with error-resolving agents and multi-turn reasoning; achieved 98% accuracy, 65% reduction in manual revision time, sub-8s latency per section across hundreds of documents monthly	
Microsoft Research	Bangalore, India
Research Intern (LLMs & AI4Code)	Jan 2024 - Jul 2024
◦ Built PwR Studio: NL→DSL translation system enforcing symbolic compilation checks on LLM outputs; reduced hallucinations by 85%, valid code generation 65%→95%; featured in Satya Nadella's keynote, adopted by 15+ organizations ◦ Architected Jugalbandi platform converting DSL into FSM-based chatbot applications; 80% reduction in development time; enabled government agencies and NGOs to deploy multilingual voice/text bots across WhatsApp, Telegram, and web; adopted by Bhashini for national chatbot infrastructure ◦ Open-sourced: PwR-Studio, PwR-NL2DSL, Jugalbandi Studio Engine	
AI Institute, University of South Carolina	Remote
Research Intern (Neuro-Symbolic AI & NLP)	Dec 2022 - Apr 2024
◦ Master's thesis with Dr. Amit Sheth: built Multimodal Ingredient Substitution Knowledge Graph (MISKG) — 27K ingredients, 80K substitution pairs, sourced from FlavorDB, ConceptNet, and Recipe1M; largest ingredient substitution corpus; adopted at UC Irvine + Stanford Health Hackathon 2024 ◦ Formulated cross-modal recipe retrieval and cooking action recognition; 95% recall; led to Cook-Gen (SMC'23)	
University of Maryland Baltimore	Remote
Research Intern (Natural Language Processing)	Oct 2022 - Apr 2023
◦ Developed personalized response generation models with Prof. Manas Gaur using reward scaling over BART & T5; improved NUBIA score by 10%; led to K-PERM (AAAI Spring Symposium 2024)	

SKILLS	
General	Python, C/C++, SQL, Docker, Kafka, Git/GitHub, AWS, Selenium, BeautifulSoup
Web Technologies	HTML/CSS, Flask, JavaScript, Beego/Gin (Golang)
Deep Learning & ML	PyTorch, TensorFlow, vLLM, OpenCV, TensorRT, SpaCy
Generative AI	Large Language Models, RAG, LlamaIndex, LangChain, Agentic Systems